Unit 5: Pragmatic & Discourse Processing

5.1 Pragmatic & Discourse Processing

1. Introduction to Discourse Processing

Discourse processing is the study of how computers understand and interpret language beyond individual sentences. It focuses on how sentences are connected and how their meaning depends on the surrounding context.

For example:

Rahul bought a new laptop. **He** is very happy with it.

To understand **"He"**, the system must determine that it refers to **Rahul**. Similarly, **"it"** refers to the laptop. This type of understanding is an important part of discourse processing.

Main objectives of discourse processing

- Identify relationships between sentences.
- Resolve references such as **he, she, it, this, that**.
- Understand the topic and context of a conversation.
- Maintain coherence between sentences.
- Identify entities and their relationships.
- Understand implied meanings.
- Track information throughout a document or conversation.

2. Pragmatic Processing

Pragmatics deals with understanding the intended meaning of language based on **context**, rather than only the literal meaning of words.

Example

Person A: *"Can you open the window?"*

Literally, this is a question about ability. However, pragmatically, it is usually understood as a **request to open the window**.

Importance of pragmatics

Pragmatic processing helps NLP systems understand:

- Speaker intention
- Context
- Implied meaning
- Conversation structure
- Politeness
- Ambiguity
- References to previously mentioned information

Applications

- Chatbots
- Virtual assistants
- Machine translation
- Question-answering systems
- Sentiment analysis
- Conversational AI
- Information extraction

3. Reference Resolution

Reference resolution is the process of identifying the entity or object to which a referring expression refers.

Example

Priya went to the library. **She** borrowed a book.

Here:

- **She** → Priya
- Priya → **antecedent**
- She → **referring expression**

The system has to determine that **"She" refers to "Priya."**

Important terms

Term	Meaning
Referring expression	Word/phrase used to refer to something
Antecedent	Entity referred to by another expression
Anaphor	Expression whose interpretation depends on previous context
Coreference	When two expressions refer to the same entity
Mention	A textual occurrence of an entity

4. Reference Phenomena

Reference phenomena describe different ways language refers to entities.

A. Anaphora

An **anaphoric reference** occurs when an expression refers to something mentioned earlier.

Example:

John bought a car. **He** likes it.

- He → John
- it → car

The earlier expression is called the **antecedent**.

B. Cataphora

Cataphora occurs when the referring expression appears **before** its antecedent.

Example:

When he arrived home, John went to sleep.

Here:

- He → John

The antecedent appears later in the sentence.

C. Exophora

Exophoric reference refers to something outside the text, usually something available from the physical or conversational context.

Example:

Put that over there.

The listener needs to observe the surrounding situation to know what **"that"** and **"there"** mean.

D. Demonstrative Reference

Words such as:

- this
- that
- these
- those

can refer to objects, events, or ideas.

Example:

The company launched a new product. **This** received positive feedback.

Here, **"This"** refers to the product launch or the event, depending on context.

E. Definite Reference

A noun phrase beginning with **"the"** can refer to an already identifiable entity.

Example:

I bought a book yesterday. **The book** is very interesting.

F. Pronoun Reference

Pronouns refer to previously introduced entities.

Examples:

- he
- she
- it
- they
- him
- her
- them

Example:

Aisha called Sara because **she** needed help.

This sentence is ambiguous because **"she"** could refer to Aisha or Sara.

5. Syntactic Constraints on Coherence

Syntactic constraints are grammatical rules that help determine whether a reference is possible.

They include:

A. Number agreement

A pronoun should generally agree with its antecedent in number.

The students finished **their** work. ✓

The students finished **his** work. ✗

B. Gender agreement

In languages or contexts where gender is grammatically marked, the reference should be compatible.

Riya said **she** was tired. ✓

C. Person agreement

Pronouns must normally maintain appropriate grammatical person.

I completed **my** work. ✓

D. Binding constraints

Certain pronouns and noun phrases have restrictions on how they can refer to other expressions within a sentence.

Example:

John saw **himself** in the mirror.

The reflexive **himself** can refer to John in this structure.

6. Semantic Constraints on Coherence

Semantic constraints use the meaning of words and events to determine whether a reference is reasonable.

Example

The boy ate the food because **he** was hungry.

“He” can naturally refer to the boy because a boy can be hungry and eat food.

Consider:

The table ate the food because **he** was hungry.

This is semantically unusual because a table normally cannot eat.

Major semantic constraints

- **Animacy:** person/animal vs. object
- **Gender**
- **Number**
- **Selectional restrictions**
- **Semantic compatibility**
- **Event relationships**
- **World knowledge**

7. Coherence in Discourse

Coherence means that different parts of a discourse are logically and meaningfully connected.

Example of coherent discourse

Rahul was hungry. He went to a restaurant. He ordered a pizza.

The sentences are connected through:

Rahul → hunger → restaurant → food

Example of weak coherence

Rahul was hungry. The computer has 16 GB RAM. The weather is cloudy.

The sentences do not have an obvious meaningful relationship.

Factors supporting coherence

1. Reference
2. Topic continuity
3. Temporal relationships
4. Causal relationships
5. Logical relationships
6. Connectives
7. World knowledge

8. Anaphora Resolution

Anaphora resolution is the task of identifying the antecedent of an anaphoric expression.

Example

Ravi went to the market because **he** needed vegetables.

The system must resolve:

he → Ravi

General steps

1. Detect the anaphoric expression.
2. Identify possible antecedents.
3. Apply grammatical constraints.
4. Apply semantic constraints.
5. Consider discourse context.
6. Select the most appropriate antecedent.

9. Hobbs Algorithm for Anaphora Resolution

The **Hobbs algorithm** is a syntactic approach to resolving **pronoun references**. It uses the **parse tree** of a sentence and searches for possible antecedents in a systematic order.

Basic idea

The algorithm:

1. Starts from the pronoun.
2. Moves upward through the syntactic parse tree.
3. Searches appropriate noun phrases.
4. Checks whether candidates satisfy constraints.
5. Selects the first suitable candidate.

Example

John saw Bill when **he** entered the room.

A syntactic parse tree is created.

The algorithm examines possible noun phrases such as:

- John
- Bill

It then uses syntactic structure and agreement constraints to identify possible antecedents.

Simplified Hobbs Algorithm

Step 1: Find the NP containing the pronoun.

Step 2: Find the lowest NP or S node that dominates the pronoun.

Step 3: Move upward in the parse tree.

Step 4: Search left-to-right through appropriate branches.

Step 5: Collect noun phrases as candidate antecedents.

Step 6: Check agreement and other constraints.

Step 7: Select a suitable antecedent.

Advantages

- Relatively simple.
- Uses syntactic structure.
- Deterministic.
- Does not require extensive training data.
- Useful as a baseline method.

Limitations

- Mainly syntactic.
- May fail when world knowledge is required.
- Can struggle with ambiguous references.
- Does not always capture discourse-level preferences.

10. Centering Algorithm for Anaphora Resolution

The topic is commonly called **Centering Theory/Centering Algorithm**.

Centering Theory models how attention moves from one discourse segment to another. It assumes that a discourse tends to maintain focus on certain entities.

Important concepts

Cb — Backward-looking center

The entity that connects the current utterance to the previous discourse.

Cf — Forward-looking centers

Entities mentioned in the current utterance that could become the focus of the next utterance.

Cp — Preferred center

The highest-ranked forward-looking center and the entity expected to be the next focus.

Example

John went to the store. He bought some milk.

First sentence:

Cf: John, store

Second sentence:

Cb: John

Because **John** continues as the main focus, the discourse is highly coherent.

Main idea

If a discourse continues to talk about the same entity, the system gives that entity a high preference when resolving pronouns.

Advantages

- Uses discourse context.
- Models topic continuity.
- Useful for pronoun resolution.

- Helps explain why some references sound more natural than others.

Limitations

- Requires discourse structure.
- Ranking can depend on grammatical roles and language.
- Ambiguous cases may still require semantic/world knowledge.

11. Hobbs vs. Centering Algorithm

Feature	Hobbs Algorithm	Centering Algorithm
Main focus	Syntax	Discourse focus
Main resource	Parse tree	Discourse context
Resolution	Candidate noun phrases	Center/focus continuity
Context usage	Limited	Strong
Complexity	Relatively simple	More discourse-oriented
Main strength	Syntactic structure	Topic continuity
Main limitation	Weak discourse modeling	Requires discourse information

5.2 Self-Learning Topics

12. Discourse Segmentation

Discourse segmentation is the process of dividing a document or conversation into meaningful units or segments.

A segment may represent:

- A topic
- An event
- A dialogue turn
- An argument
- A paragraph
- A discourse relation

Example

Segment 1: The company introduced a new AI product.
Segment 2: The product received positive customer feedback.
Segment 3: As a result, the company increased production.

The three parts form related but distinguishable discourse segments.

Why segmentation is important

Discourse segmentation helps NLP systems:

- Identify topic changes.

- Understand document structure.
- Detect relationships between sentences.
- Improve summarization.
- Improve question answering.
- Analyze conversations.
- Perform information extraction.

Common segmentation clues

1. Paragraph boundaries
2. Topic changes
3. Discourse connectives
4. Changes in participants
5. Changes in time
6. Changes in location
7. Changes in events
8. Linguistic cues

Examples of discourse connectives

- however
- therefore
- because
- although
- meanwhile
- furthermore
- in addition
- as a result

13. Conference Resolution

The term **"Conference resolution"** is generally understood in NLP as **coreference resolution**. If your syllabus specifically uses "Conference resolution," it may be a spelling/wording variation.

Coreference Resolution

Coreference resolution is the process of determining which expressions in a text refer to the **same real-world entity**.

Example

Amit bought a new phone. **He** uses **it** every day.

Coreference relationships:

- Amit ↔ He
- new phone ↔ it

Difference between anaphora resolution and coreference resolution

Anaphora Resolution	Coreference Resolution

Identifies an expression's antecedent	Identifies expressions referring to the same entity
Often focuses on pronouns	Covers pronouns, names, noun phrases, etc.
Usually discourse-dependent	Broader entity-linking task
Example: He → Rahul	Rahul ↔ He ↔ the student

Example of coreference chain

Dr. Sharma joined the university. **He** became the head of the department. **The professor** later retired.

Possible coreference chain:

Dr. Sharma → He → The professor

14. Applications of Discourse and Reference Processing

These techniques are widely used in:

- **Chatbots:** maintaining conversational context
- **Virtual assistants:** understanding pronouns and references
- **Machine translation:** maintaining correct references across sentences
- **Text summarization:** identifying important entities and topics
- **Information extraction:** connecting facts to entities
- **Question answering:** understanding what “he,” “it,” or “they” refers to
- **Document analysis:** tracking entities throughout documents
- **Search engines:** improving contextual understanding

15. Quick Revision Points

Pragmatics

- Deals with meaning in context.
- Understands speaker intention and implied meaning.

Discourse

- Deals with connected language beyond individual sentences.
- Focuses on coherence and relationships between sentences.

Reference Resolution

- Determines what a referring expression refers to.

Reference Phenomena

- Anaphora
- Cataphora
- Exophora
- Demonstrative reference

- Definite reference

Coherence

- **Syntactic constraints:** grammar, number, gender, binding.
- **Semantic constraints:** meaning, animacy, world knowledge, compatibility.

Hobbs Algorithm

- Syntax-based.
- Uses parse trees.
- Searches for suitable antecedents.

Centering Algorithm

- Discourse-based.
- Tracks discourse focus.
- Uses **Cb, Cf, and Cp**.

Discourse Segmentation

- Divides discourse into meaningful units.
- Useful for topic and structure analysis.

Coreference Resolution

- Identifies expressions referring to the same entity.
- Example: **“Ravi → He → the student.”**